

Miniature Capacitance Level Switch For Liquid Application Model: MCLS-900LS



The MCLS-900LS is a compact and reliable capacitance-type point level switch designed for accurate point level detection of liquids and solids in tanks, vessels, hoppers and process equipment. Its miniature construction makes it ideal for applications where compact size, easy installation and dependable level switching are required. The instrument detects the presence or absence of material at a predetermined level by sensing the change in capacitance between the probe and the surrounding vessel or reference ground. When the material reaches the sensing probe, the change in capacitance is detected by the electronic circuit, which generates a switching output for high-level or low-level indication and process control.

SALIENT FEATURES:



- Simple Connections – Easy and quick electrical installation.
- PNP Output – Suitable for direct interfacing with PLC and automation systems.
- Compact Design – Suitable for installations with limited space.
- Point Level Detection – Reliable high-level and low-level detection.
- Light & Powdered Materials – Optimized for lightweight and powdered bulk solids.
- Wear-Free Operation – No moving parts, ensuring long operating life.
- Maintenance-Free Design – Minimal maintenance requirements.
- Accurate Switching Function – Reliable switching independent of normal process variations.
- Build-Up Tolerance – Designed to minimize the effect of material build-up on the sensing probe.
- Multiple Mounting Options – Available with 1/2" BSP(M), 1/2" NPT(M) and other process connections.
- Reliable Performance – Suitable for continuous industrial point-level monitoring.

APPLICATIONS:

The MCLS-900LS is a compact capacitance-type point level switch suitable for reliable level detection of fine-grained, powdered bulk solids and liquid media. Suitable for point-level detection of a wide range of liquids, including water, oils, chemicals, solvents, resins, adhesives, emulsions and various industrial process liquids.

Miniature Capacitance Level Switch For Liquid Application Model: MCLS-900LS

OPERATING PRINCIPLE:

An AC electrical signal is generated at the tip of the sensing electrode. When the sensor is covered by the process medium, the capacitance at the sensing probe changes. This change is detected by the electronic circuitry and converted into a reliable switching signal for point-level detection. the instrument is designed to tolerate a certain degree of material build-up on the sensing probe. this helps minimize the influence of build-up on the switching function and ensures stable and reliable level detection under normal process conditions.

COMPONENTS OF COMPACT LEVEL SWITCH ARE EXPLAINED IN FIGURE



- -----Cable Entry DIN Connector
- -----Acrylic
- -----SS Body
- -----Mounting Thread
- ----- Sence

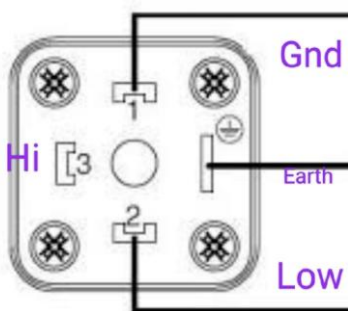
RELAY OUTPUT ELECTRICAL CONNECTION Supply: 12 VDC To 30 VDC



ELECTRICAL CONNECTION

POWER SUPPLY: 12 VDC TO 30 VDC

OUTPUT: PNP NO



Miniature Capacitance Level Switch

For Liquid Application Model: MCLS-900LS

DATA SHEET

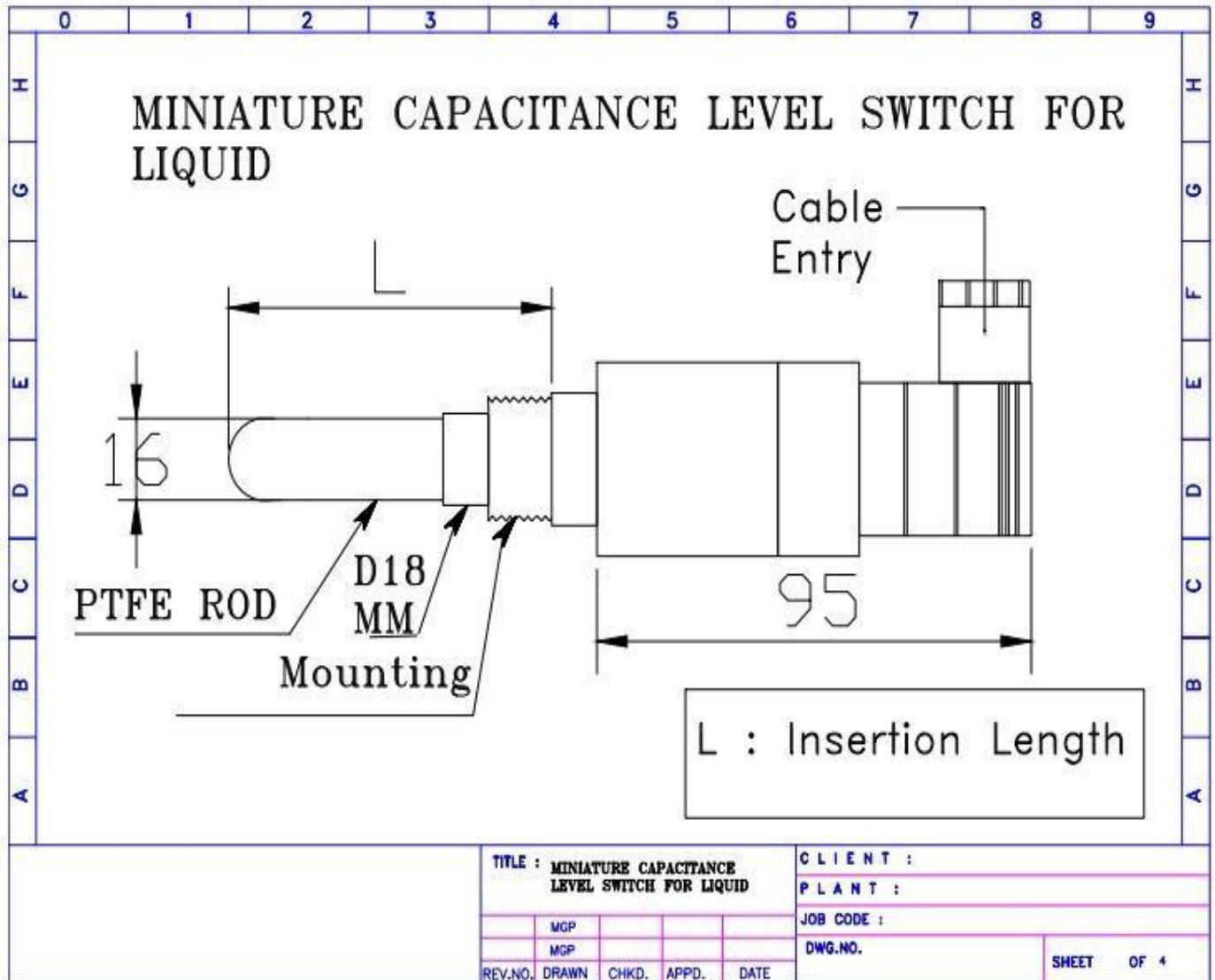
MCLS-900LS– Miniature Capacitance Level Switch

Selection / Ordering Information

Selection	Code	Description
Model No.	MCLS-900LS	Miniature Capacitance Level Switch
Application	LS	Liquid Solid
Mounting Type	IN	Integral
Insulation	FP	Full PTFE Insulation
Probe Material	04 /4L/ 16/ 6L/ SM	SS304/SS304L / SS316/ SS316L / Special material
Power Supply	DC	24VDC
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED
Enclosure	AH/ WP / FL	*Acrylic /Weatherproof / Flameproof
Connection	T	Threaded
	F	Flanged
Mounting	B0/B1 / B2 / B3/B4	½"/1" / 1½" / 2" BSP/ Special Mounting
	N0/N1 / N2 / N3/N4	½"/1" / 1½" / 2" NPT/ Special Mounting
Flange Rating	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
Flange Facing	RF / FF	Raised Face / Flat Face
MOC-Process Connection	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
Flange Size	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
Pressure	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customized
Temperature	T1/ T2/TS	Max -20–150°C / 200°C/Customer Specified Special Designed
High Temperature stand off	H8/HS	80MM/Special standoff Extension
Cable Gland	G0/G1/ G2 / G3/ G4	PG 9/ PG13.5 / ½"NPT / ¾" ET/
Ground / Reference	GS / GX	Standard / Special
Material Certificate	M0 / MC	Not Required / Material test certificate
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	XXXX	40 mm TO Customized

Miniature Capacitance Level Switch For Liquid Application Model: MCLS-900LS

Mechanical Drawing:



Manufactured By:

Levtron Instruments Pvt Ltd.

Office No 2, 1st floor, plot no 6, Pandhari Industrial estate, Shivane. Pune 411023, Contact: +91 9579430012/+91 9730042725, E-mail: sales@levtron.in
www.Levtron.in/ www.Levtroninstruments.com/www.levelsence.in